

Ganesh Padakanti

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SUMMARY

Experienced Mechanical Design Engineer with 5+ years in electro-mechanical product design, CAD modeling, tolerance-controlled assemblies, and manufacturing verification. Proficient in SolidWorks, Fusion 360, and prototype validation with a strong focus on design clarity, manufacturability, and cross-disciplinary team collaboration, ready to deliver robust mechanical systems and support end-to-end product development.

EXPERIENCE

MECHANICAL DESIGN ENGINEER

Graviti Pharmaceuticals Pvt Ltd (**May 2024 - Present**)

- Developed standardized 2D manufacturing drawing templates with GD&T to improve clarity, repeatability, and manufacturing interpretation across the design team.
- Performed tolerance analysis and part-level fit validation for PCB, sensors, and micro-components within compact electromechanical assemblies.
- Contributed to electro-mechanical subsystem design for compact product modules with strict functional requirements.
- Supported design verification and test execution through prototype evaluation and performance tracking.
- Applied DFMA principles for CNC machining, acrylic fabrication, and injection-molded parts in coordination with vendors.
- Led multiple SLA/FDM prototype iterations to validate assembly feasibility, structural integrity, and thermal behavior.
- Worked with test engineering to verify mechanical performance and document results for design sign-off.
- Troubleshoot performance, fit, and system integration issues during prototype validation cycles and implemented corrective design improvements.
- Improved design release workflow through structured CAD practices, reducing drawing errors and manual rework.
- Reduced prototype cycle time by 30% through faster design iterations and cross-functional reviews.
- Ensured CAD, drawing, and BOM data consistency during production builds through close vendor coordination.
- Collaborated with electrical, hardware test, and manufacturing teams to align mechanical solutions with system requirements.

PRODUCT DESIGN ENGINEER

Refroid Technologies Pvt Ltd (**Sep 2021 -Nov 2023**)

- Created detailed 2D manufacturing drawings with GD&T for fabricated brackets, mounting systems, and structural assemblies to ensure fabrication and assembly accuracy.
- Standardized drawing structure and CAD practices to reduce manufacturing queries and improve documentation quality.
- Performed structural and thermal simulations using SimSolid to validate rigidity and load-bearing capacity of assemblies.
- Collaborated with electrical and electronics teams to align mechanical structures with component constraints and routing requirements.
- Worked closely with manufacturing vendors to validate feasibility of welded, bent, and machined components during design stages.

- Reduced part count by 15% through modular structural design, improving assembly efficiency and serviceability.
- Conducted material selection and structural optimization resulting in 10% manufacturing cost reduction.
- Participated in prototype testing and performance evaluation to improve mechanical reliability.

PRODUCT DESIGN INTERN

Artenal- Vancouver, CA (**Jan 2021 – July 2021**)

- Contributed to the mechanical design of gripper mechanisms and chassis structures for an agricultural robotic fruit harvester using Fusion 360.
 - Modeled functional subassemblies considering actuation space, structural strength, and assembly feasibility.
 - Supported concept development through CAD iterations and early-stage design validation.
 - Participated in early-stage concept ideation and mechanical problem-solving sessions.
 - Assisted in prototype feasibility studies focusing on manufacturability and structural stability.
 - Collaborated with a remote cross-functional team during design and development phases.
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SOFT SKILLS

- Strong analytical approach to assembly and structural design problems
- Cross-team communication with testing, electrical, and validation teams
- Problem solving in integrated hardware and prototype environments
- High ownership of manufacturing-ready documentation and CAD quality
- Detail-oriented with focus on tolerances, fit, and assembly feasibility

TECHNICAL SKILLS

- **CAD Tools:** SolidWorks, Fusion 360, Creo, AutoCAD
 - **Design Practices:** DFMA, GD&T, tolerance analysis, manufacturing drawings, concept development
 - **Prototyping:** FDM/SLA 3D printing, CNC machining, acrylic fabrication
 - **Validation:** Assembly testing, prototype validation, structural & thermal simulation (SimSolid)
 - **CAD Standardization:** Drawing templates, BOM alignment, design workflow improvement
 - **Programming (Basic):** Python, CAD macros for template automation
 - **Mechanical System Validation:** Prototype test execution, performance evaluation
 - **Design Verification:** Test support documentation, issue tracking, design refinement
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EDUCATION

- **Bachelor of Technology (B.Tech)** – Mechanical Engineering
Swami Vivekananda Institute of Technology, Hyderabad – 2020

LANGUAGES

- English | Hindi | Telugu | German (A1)